JC TECH KNOWLEDGE HUB -- TECHNICAL SPECIFICATION SERIES

EV Component Database

Electric Powertrain Architecture, Subsystem Specifications & Sourcing Matrix

1. Electric Vehicle Architecture & Systems Integration

The electrification of passenger, commercial, and two/three-wheeler mobility requires an intricate synthesis of high-voltage electrochemical storage, high-frequency power electronics, electromechanical energy conversion, and sophisticated thermal management. The JC Tech EV Component Database catalogues commercially available and production-proven components, providing mechanical engineers, powertrain integrators, and sourcing specialists with verified technical parameters, form-factors, cooling requirements, and safety compliance benchmarks.

2. Powertrain Component Classification Matrix

- **Traction Battery Systems:** High-energy and high-power cells (Prismatic, Cylindrical 21700/4680, Pouch), series/parallel modular assemblies, liquid cold-plates, and Battery Management Systems (BMS) with ISO 26262 ASIL-D functional safety.
- **Electric Traction Motors:** Permanent Magnet Synchronous Motors (PMSM), Induction Motors (IM), and Switched Reluctance Motors (SRM). Features continuous/peak torque maps, field-weakening capabilities, and hollow-shaft direct rotor cooling.
- **Inverters & Power Electronics:** Silicon Carbide (SiC) and Silicon IGBT traction inverters (400V / 800V architectures), bidirectional DC-DC converters (HV-to-LV 12V/24V), and On-Board Chargers (OBC) with integrated V2L/V2G capabilities.
- **High-Voltage Distribution & Protection:** Power Distribution Units (PDU), pyrotechnic battery disconnect fuses, high-voltage interlock loops (HVIL), shielded cable harnesses, and hermetically sealed DC contactors.

3. Lithium-Ion Battery Chemistries Comparison

Chemistry	Nominal Cell Volts	Energy Density	Cycle Life (80% DoD)	Thermal Stability & Safety Profile
LFP (LiFePO4)	3.2 V	120 - 165 Wh/kg	3,000 - 6,000+	Exceptional; thermal runaway threshold > 270 deg C; zero fire risk
NMC 811 (Ni-Mn-Co)	3.6 - 3.7 V	240 - 280 Wh/kg	1,200 - 2,000	Moderate; thermal runaway starts ~210 deg C; requires active BTMS
NCA (Ni-Co-Al)	3.6 V	250 - 300 Wh/kg	1,000 - 1,500	Moderate; high energy density; requires precise cell temperature balancing
Sodium-Ion (Na-Ion)	3.0 - 3.1 V	130 - 160 Wh/kg	2,500 - 4,000	High; safe at zero volts for shipping; superior low-temp discharge
Solid-State (Emerging)	3.8 V	350 - 450 Wh/kg	1,500 - 2,500	Superior; non-flammable solid electrolyte; elimination of dendrite shorts

4. Deep Dive Specification: 800V SiC Traction Inverter

Engineering Parameter	Specification Value	Design Notes & Testing Standard
DC Bus Voltage Range	500 V - 900 V (800V Nominal)	Enables ultra-fast DC charging (10% to 80% in <18 minutes)
Peak Power / Continuous	220 kW Peak / 120 kW Continuous	Liquid cooled with 50/50 water-ethylene glycol at 15 L/min
Power Semiconductor Tech	Silicon Carbide (SiC) MOSFET Modules	Lower switching losses (<30% vs IGBT); allows >20 kHz PWM switching
Peak Inverter Efficiency	99.2% at nominal operating load	Significantly extends vehicle cruising range by 4% to 7%
Communication & Safety	CAN-FD / Auto-Ethernet (ASIL-D)	ISO 26262 compliant fault detection, hardware overcurrent trip <2 us
Environmental Protection	IP67 / IP6K9K Hermetic Enclosure	Tested per ISO 16750 for automotive vibration, salt spray, shock

5. Powertrain Sizing Analytical Formulas

- **Total Tractive Effort (F_total):** F_total = F_rolling + F_aerodynamic + F_gradient + F_acceleration. Sized for 0-100 km/h sprint and 30% hill grade climb.
- **Aerodynamic Drag Force:** F_aero = 0.5 * rho_air * C_d * A_frontal * v^2. Dictates highway speed power draw and pack kWh sizing.
- **Required Motor Shaft Power:** P_motor = (F_total * v) / (eta_transmission * eta_motor). Governs continuous thermal power rating.

6. Key Homologation & Regulatory Compliance Standards

AIS-156 / 038
Indian EV Safety Standard

UN ECE R100
Global High-Voltage Safety

ISO 26262
ASIL-D Functional Safety

IEC 61851
Conductive Charging Norms

Request Full CAD Models & Supplier Sourcing Intelligence
The full JC Tech EV Database includes STEP models of motors and inverters, thermal simulation flow rates, and verified supplier contact lists across India, Europe, and East Asia.